

Logarithmic functions



1 Consider the function $y = \log_2 x$.

a Complete the table of values.

x	$\frac{1}{2}$	1	2	4	16
y					

b Sketch a graph of the function.

c State the equation of the vertical asymptote.

d State whether the following elements are key features of the graph of $y = \log_2 x$:

i A y -intercept

ii A horizontal asymptote

iii An x -intercept

iv A lower limiting value

v An upper limiting value

2 Sketch the graph of $y = \log_5 x$.

3 Consider the function $y = \log_3 x$.

a Complete the table of values.

x	$\frac{1}{243}$	$\frac{1}{3}$	1	3	9	81
y						

b Is $y = \log_3 x$ an increasing or decreasing function?

c Describe the behavior of $y = \log_4 x$ as x approaches 0.

d State the value of y when $x = 0$.



Inverses of logarithmic functions

4 Consider the function $y = \log_2 x$.

a Complete the table of values for $y = \log_2 x$:

x	1	2	4	8
y				

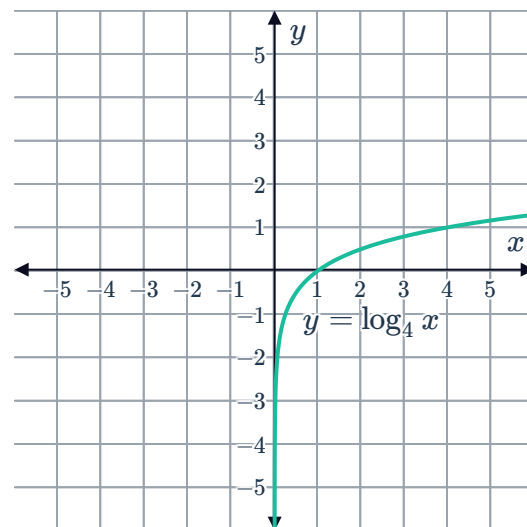
b Complete the table of values for the inverse function of $y = \log_2 x$:

x				
y				

c Sketch the graph of $y = \log_2 x$ and its inverse function on the same coordinate plane.

d State the equation of the inverse function of $y = \log_2 x$.

5 Consider the graph of $y = \log_4 x$.
Sketch the graph of the inverse of $y = \log_4 x$.



6 For each of the following functions:

i Graph $f(x)$, the line $y = x$ and the inverse of $f(x)$ on the same coordinate plane.

ii State the equation of the inverse function.

a $f(x) = 4^x$

b $f(x) = \left(\frac{1}{4}\right)^x$

c $f(x) = \left(\frac{1}{5}\right)^x$

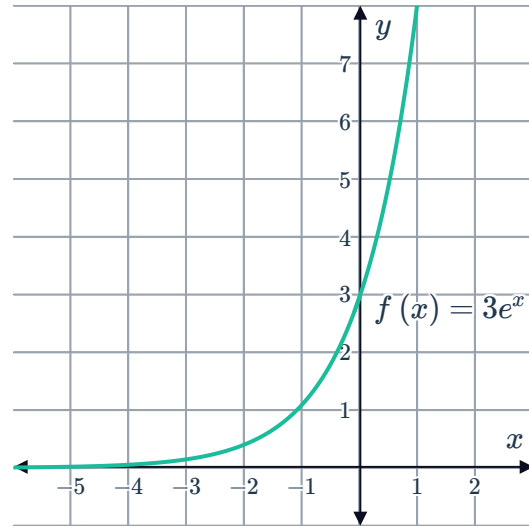
Natural logarithmic functions



7 Describe the logarithmic function known as $\ln x$ in terms of its base.

8 Consider the graph of $f(x) = 3e^x$:

- a Sketch the graph of the inverse of $f(x)$.
- b State the equation of the inverse function.



9 Consider the function $f(x) = 2e^x - 5$ for all $x \in \mathbb{R}$.

- a State the equation of the inverse function.
- b Using a graphing calculator, state the coordinates of the points at which the graphs of $y = f(x)$ and $y = f^{-1}(x)$ intersect. Round all values to three decimal places.

Transformations of logarithmic functions

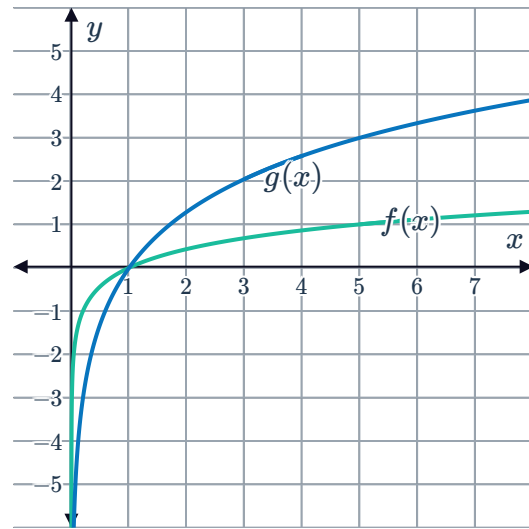
10 If the function $f(x) = \log_3 x$ is translated 5 units to the right, state the equation of the resulting function.

11 Sketch the graph of $y = \log_3 x$ after it has been translated 4 units down.

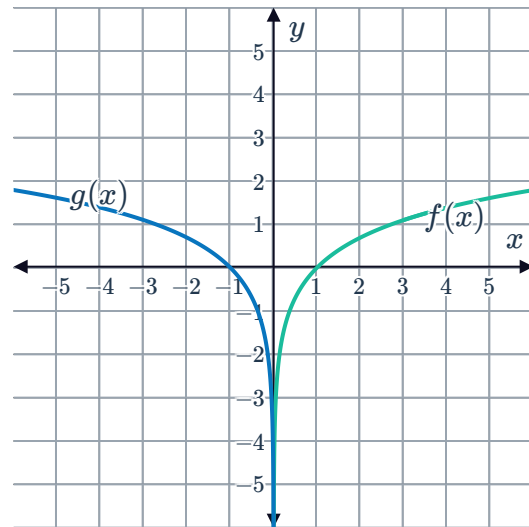
12 For each of the following functions:

- i State the equation of the function after it has been translated.
 - ii Sketch the translated graph.
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- a $y = \log_5 x$ translated downwards by 2 units.
 - b $y = \log_3(-x)$ translated upwards by 2 units.

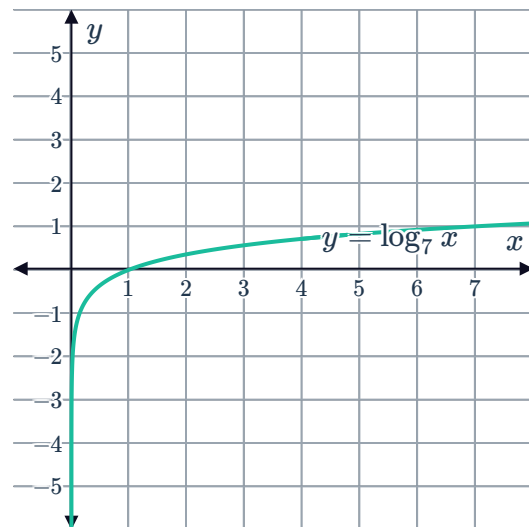
- 13 Consider the graphs of $f(x) = \log_5 x$ and $g(x)$:
Find the equation of $g(x)$.



- 14 Consider the graphs of $f(x) = \ln x$ and $g(x)$:
- Describe the transformation that has been performed on the graph of $f(x) = \ln x$ to obtain $g(x)$.
 - State the equation for $g(x)$.



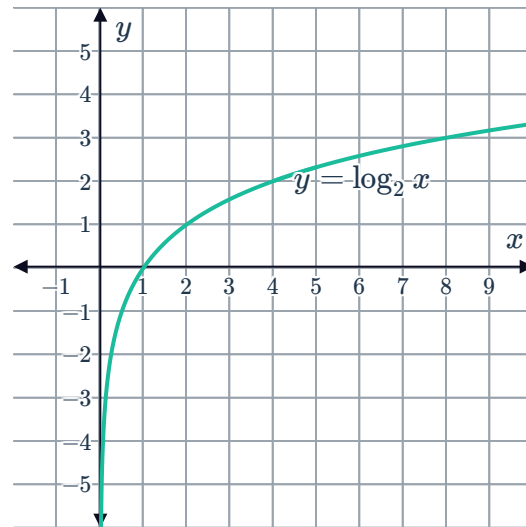
- 15 Consider the graph of $y = \log_7 x$:
- Describe the transformation needed to obtain the graph of $y = -3 \log_7 x$ from $y = \log_7 x$.
 - Sketch the graph of $y = -3 \log_7 x$ on the same coordinate plane as $y = \log_7 x$.



- 16 Given the graph of $y = \log_2 x$, sketch the graph of the following functions:



a $y = \frac{1}{3} \log_2 x$ b $y = -\frac{1}{2} \log_2 x$

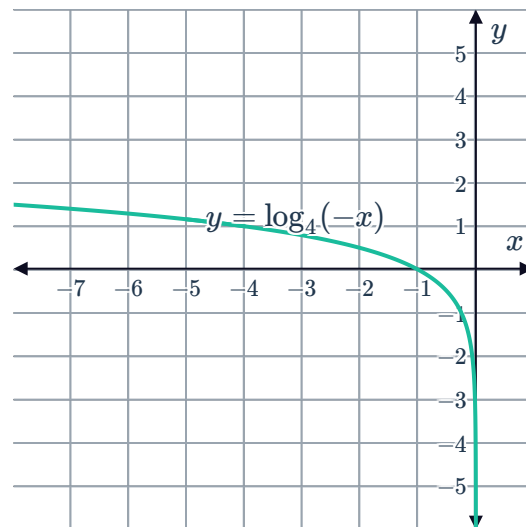


- 17 Consider the graph of the function $y = \log_4 (-x)$:

- a Complete the table of values for $y = \log_4 (-x)$:

x	-4	-1	$-\frac{1}{4}$
$\log_4 (-x)$			

The graph of $y = \log_4 (2 - x)$ can be obtained from the original graph by transforming it.



- b Complete the table of values for $y = \log_4 (2 - x)$:

x	-2	1	$\frac{7}{4}$
$\log_4 (2 - x)$			

- c Describe the transformation from $y = \log_4 (-x)$ to $y = \log_4 (2 - x)$.

- 18 Consider the function $f(x) = \log_3 (x - 4)$.

- a State the equation of the vertical asymptote of $f(x)$.
b State the coordinates of the x -intercept of the function.
c Determine the exact value of $f(7)$.
d Sketch a graph of $f(x) = \log_3 (x - 4)$.

19 Consider the function $f(x) = \log_3(2 - x)$.



- a State the equation of the vertical asymptote of $f(x)$.
- b State the coordinates of the x -intercept of the function.
- c Determine the exact value of $f(-7)$.
- d Sketch a graph of $f(x) = \log_3(2 - x)$.

20 The graph of $y = \log_4 x$ has a vertical asymptote at $x = 0$. By considering the transformations that have taken place, state the equation of the vertical asymptote of the following functions:

- | | |
|---------------------------|------------------------|
| a $y = 2 \log_4 x - 4$ | b $y = 2 \log_4 x$ |
| c $y = \log_4(x - 5)$ | d $y = -\log_4 x$ |
| e $y = \log_4(x + 3) - 2$ | f $y = 3 \log_4 x + 2$ |

21 For each of the following functions:

- i Solve for the x -intercept.
- ii State the equation of the vertical asymptote.
- iii Sketch the graph of the function.

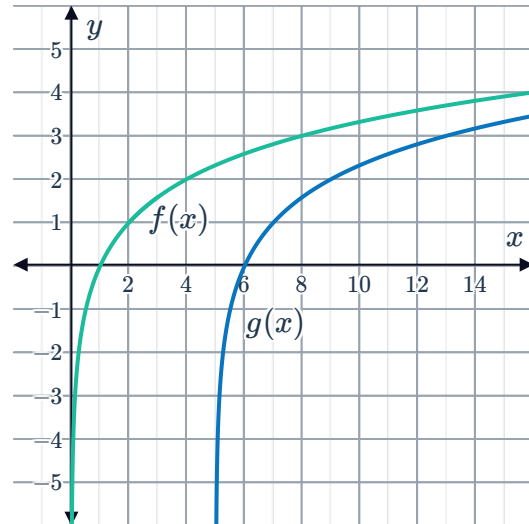
- | | |
|------------------------------|------------------------------|
| a $f(x) = 4 \log_2(x - 7)$ | b $f(x) = -\log_4(x + 4)$ |
| c $f(x) = \log_2(x - 1) - 4$ | d $y = -2 \log_3(x + 8)$ |
| e $y = -\log_4(x + 9) + 1$ | f $y = -4 \log_3(x - 1) - 4$ |
| g $y = \log_2(-x - 1) - 1$ | h $y = -\log_2(-x + 2) + 2$ |



Domain and range of logarithmic functions

22 Consider the graphs of the functions $f(x) = \log_2 x$ and $g(x) = \log_2(x - 5)$:

- a** Describe the transformation from $f(x)$ to $g(x)$.
- b** State whether the domain and range of $f(x)$ change in its transformation to $g(x)$.



23 For each of the following functions:

- i** State the equation of the inverse function.
- ii** State the domain of the inverse function using interval notation.
- iii** State the range of the inverse function using interval notation.

a $f(x) = \log_e 5x$ for all $x > 0$

b $f(x) = \log_e(4x - 3)$ for all $x > \frac{3}{4}$

c $f(x) = 4 \log_e(2x - 5)$ for all $x > \frac{5}{2}$

d $f(x) = 6 \log_e 5x - 3$ for $x > 0$

e $f(x) = 6 \log_e(x - 4) - 3$ for $x > 4$

f $f(x) = e^{x+3}$ for all $x \in \mathbb{R}$

g $f(x) = e^x - 3$ for all $x \in \mathbb{R}$

h $f(x) = 5 - e^{-x}$ for all $x \in \mathbb{R}$

i $f(x) = 2e^{x+3}$ for all $x \in \mathbb{R}$

j $f(x) = 3e^{5x} + 4$ for all $x \in \mathbb{R}$

24 State the domain of the following functions as an inequality:

a $y = \log 3x$

b $y = \log(-7x)$

c $y = \log\left(\frac{1}{2}x\right)$

d $y = \log\left(-\frac{1}{4}x\right)$

e $y = \log(3x + 9)$

f $y = \log_3(x + 5) - 4$

g $y = \log_2(x^2 - 10x + 21)$

h $y = \log(x^3 - 16x)$

i $y = 3 \log_{\frac{1}{4}}(x + 5)$

j $y = -4 \log_2(3 - x)$

k $u = \ln(x^2 + 9)$

l $u = \ln(-x^2 + 25)$

